

Classifying non-stationary time-varying signals on graphs through their deviation from a stationary behavior: preliminary results on EEG signals

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I. ABSTRACT

Brain Computer Interfaces systems in general aspect are systems that transform brain signals into control signals. Nowadays BCI has different applications with large variety of tasks such as Prosthetic Control [1], communication and motor control in locked-in patients [2], virtual reality [3], BCI typewriter [4], learning alternative neural paths to control arm movement in stroke survivors [5], diagnosing brain diseases [6] and control of devices [7], [8], [9].

In this study, we focus on electroencephalography (EEG) signals for brain signal acquisition, which are known to be high-dimensional, non-stationary and noisy.

We consider here the classification problem of finding the Motor Imagery (MI) class — *e.g.*, mental visualization of a move of a hand or a foot — associated with a given EEG signal. Hypothesis through this paper is that different parts of EEG signals have different values in extraction of MI features. And The goal of this paper is to improve classification error based on selecting a nonstationary parts of EEG data over time. Our result shows that nonstationarity filtering based on this method improves the accuracy of classification even with using of a simple classifier like SVM.

II. DATA SET AND PREPROCESSING

A. BCI competition III data set IVa

This dataset contains EEG signals from five healthy subjects performing right hand and foot motor imagery without feedback. Visual cues are shown for 3.5^{sec} to indicate that which of the following 2 motor imageries should be performed: (R) right hand, (F) foot. There are periods with random lengths 1.75^{sec} to 2.5^{sec} , between each ques' representations. Recordings of 118 EEG channels in frequency range of $[0.05^{Hz}, 200^{Hz}]$, digitized at 100^{Hz} and 1000^{Hz} , are available. For each subject 280 trials are recorded some of them have target markers for training a classifier and the rest for testing phase [10].

B. BCI competition IV data set IIa

This dataset contains EEG signals from nine healthy subjects performing left hand, right hand, both feet, and tongue motor imagery without feedback. At the beginning of each trial ($t = 0$) short acoustic warning was presented till $t = 2^{sec}$, afterward a cue was shown for 1.25^{sec} to indicate that which of the 4 motor imageries should be performed. Then the

subjects were asked to carry out the motor imagery task till $t = 6^{sec}$. Recordings of 22 EEG channels in frequency range of $[0.5^{Hz}, 100^{Hz}]$, digitized at 250^{Hz} , are available. For each subject 288 trials are recorded (72 trials per class) for training part and 288 trials for test phase. [10]. In this paper we used two classes out of four (left and right hand) for evaluation of our method.

C. Preprocessing

Before training a classifier, EEG signals are generally preprocessed. This step mostly contains frequency filtering, extracting the interesting intervals, dimensionality reduction and artifact rejection (for tackling with eye blinks, heart beat, muscular, movements and etc). In this work we use data with 100^{Hz} sampling rate that they are band pass filtered in frequency range of $[7^{Hz} - 32^{Hz}]$ using a 4th order Butterworth bandpass filter. Afterward we extract time interval between $[0.5^{sec} - 2.5^{sec}]$ after indication of ques ($t = 0^{sec}$) for dataset III data set IVa $[2.5^{sec} - 4.5^{sec}]$ for dataset IV data set IIa. The method we introduce in this article includes formulation to improve artifact rejection in itself, so we didn't use any other preprocessing method.

III. EXISTING METHODS

A. Classical Signal Processing Methods

Feature extraction methods in BCI from different perspectives has been considered. Fourier and time-frequency based features are used in [11], [12]. Time-Frequency-Spatial analysis is done in [13], [14]. Wavelets are considered in [15], [16] for detection and quantification of the ERP components that used for classification of different task. Classification based on coefficients of ARMA model has done in [17], [18].

Authors in [19] applied PCA method to short segments of data and considered the basis vectors themselves as features for classification. Hand grasping and opening and the resting state are classified in [20] based on features from ICA method. Also ICA method are used to separate the P300 source from the background noise [21].

B. CSP Method

For the aim of classifying motor imageries, the method of Common Spatial Patterns (CSP) has been shown to achieve good classification results. The main idea is to find a linear transformation w , that maximizes the variance of the one

class while minimizing the variance of the other class at the same time. After finding this weights one can use logarithm of variance of transformed data as features. At last classification methods can be applied on this features. But CSP algorithm also has some weaknesses due to noise and artifacts in the data, that causes lack of robustness. It is known that CSP could be fitted on the noise or artifacts instead of data [22]. For tackling this problem numerous methods has been proposed that modify classic CSP algorithm in different ways. One can divide them into 4 division.

- 1) **Improving the Covariance Part:** in [23] authors proposed using of neurophysiological knowledge of brain and maximizing the ratio of variances of EEG sources originating inside and outside the region of interest (ROI). In [24], [25] a method for calculating robust common spatial filters by maximizing the minimum variance ratio within a prefixed set of covariance matrices are presented. Classical CSP uses average covariance matrix over each class, but outliers could affect average estimates. So for tackling with this problem authors in [26] proposed robust estimates of the class-wise covariance matrices through solving an optimization problem.
- 2) **Regularized Methods:** Selecting the best number of electrodes in BCI system is always matter of fact. in [27], [28] authors proposed a channel selection method based on approximation of l_0 with l_1/l_2 norm. A generalization of CSP is proposed in [29]. And authors introduced several new methods for calculating CSP filters based on different regularizer functions. In [30], [31], [32] problem of between class nonstationarity is considered so outlier trials are penalized in the classical CSP problem.
- 3) **Divergence Based Methods:** In [33], [34], [35], [36] equivalent problem to classical CSP problem from information geometric point of view is introduced. In this framework classical CSP could be generalized to a broad range of BCI problems and maintains a desirable conditions. Divergence between the data distribution of individual trials and the overall data distribution could be considered as a measure of nonstationarity. In [37] this measuer is added to the main objective function of CSP for reducing the influence of artifacts and nonstationarity. in [38] problem of automatic time interval selection in EEGs is studied and a method for extracting an interval with minimum irregularity based on entropy measuring is proposed.
- 4) **Spectral filtering Methods:** As far as classical CSP doesn't consider the spectral characteristics of EEGs several methods were introduced to capture spectral behaviour of EEGs [39], [40], [41]. Learning optimum spatio-spectral filters by maximizing the mutual information between the feature vectors and the corresponding class labels is studied in [42]. Also finding spatial filters using joint characteristics of the spatial and spectral domains is considered in [43].
The main idea of CSP and its modifications is to identify one or two projection matrices that discriminant

informations of different classes in space of projected data without considering the structure of data.

C. Graph Based Methods

Because Classical approaches do not consider the structure of EEG signals, newer approaches based on graph signal processing have been developed [44], [45]. Authors In [46] proposed a dimensionality reduction and classification algorithm based on Riemannian geometry of covariance matrices. While many of these techniques assume stationarity and/or smoothness of graph signals — which does not match the signal model considered here —, some recent approaches have started to consider more complex non-stationary signals on graphs [47], [48], [49], [50], [51], [52]. Still, none of these methods assume signals to be both non-stationary on graphs, and time-varying. More recently MI classification from deep learning point of view is analyzed and results show that this method with comparison to other classical methods achieves better result [53].

In this work, we propose to discuss a possible *regularization* of non-stationarity, by studying the deviation of an EEG signal from its stationary behavior on a graph that we model by the covariance matrix of the electrodes. The postulate of this article is that signals from different classes should deviate from their stationary behavior in distinct manners. We show here our first observations on this methodology.

IV. METHODOLOGY

Let $\mathcal{X}$ be our set of EEG signals, each element of the set being a $N \times T$ matrix of measurements of N -dimensional signals for T consecutive time samples.

Let $\mathcal{C}$ be the MI classes associated with the elements in $\mathcal{X}$. When considering a two-class problem, we have $\forall c \in \mathcal{C} : c \in \{\text{hand}, \text{foot}\}$.

Let us consider an EEG signal $\mathbf{X} \in \mathcal{X}$. We know from [54] that the covariance matrix of stationary signals on graphs has the same eigenvectors as a matrix representing the graph on which they evolve. We choose to analyze the deviation of $\mathbf{X}$ to its stationary behavior. Since neurons are more organized during the MI task interval the idea is we should have nonstationarity parts when some MI happens. We measure nonstationarity over time as follows:

- 1) We extract submatrices of $\mathbf{X}$ using a time-varying sliding window, so that we obtain a series $(\mathbf{X}^{1:W}, \mathbf{X}^{S+1:W+S}, \dots, \mathbf{X}^{T-W+1:T})$, where W_t is the time window size and S is the stride. Each of these elements is a matrix of dimensions $N \times W$.
- 2) For each of the elements $(\mathbf{X}^{jS+1:jS+W}, j = 0, 1, \dots)$ of this series, we compute its $N \times N$ sample covariance matrix $\tilde{\Sigma}^j$.
- 3) We compute the overall sample covariance matrix $\tilde{\Sigma}$.
- 4) We compute the projections $(\tilde{\mathbf{A}}^j)_j$ of the matrices $(\tilde{\Sigma}^j)_j$ in the eigenbasis of $\tilde{\Sigma}$ as follows:

$$\forall j : \tilde{\mathbf{A}}^j = \tilde{\chi}^\top \tilde{\Sigma}^j \tilde{\chi}, \quad (1)$$

where $\tilde{\chi}$ are the eigenvectors of $\tilde{\Sigma}$ (see [54] for additional details).

- 5) Because the eigenvectors of $(\tilde{\Sigma}^j)_j$ are not aligned with $\tilde{\chi}$ due to non-stationarity of signals and a limited number of samples, $(\tilde{\Lambda}^j)_j$ is not in general a diagonal matrix. And because according to Kullback-Leibler divergence the closest diagonal matrix to $\tilde{\chi}$ is its diagonal [55]. Therefore, we approximate it with its diagonal elements $\tilde{\Lambda}^j$.

By doing so, we are performing some sort of regularization to study deviation of a time signal on a graph from its stationary behavior on an unknown graph.

V. RIEMANNIAN DISTANCE (SIMPLIFIED VERSION)

In this part we used special case of Wasserstein metric which assumes $\tilde{\Sigma}^j$ are commuting (they have same eigenvectors) so Σ_t has closed form as follows [56]:

$$\Sigma_t = \left(\sum_j (\tilde{\Sigma}^j)^{\frac{1}{2}} \right)^2 \quad (2)$$

Also for characterizing non-stationary intervals we use following measure:

$$\Gamma(\tilde{\Lambda}^j, \tilde{\Lambda}^j) = \left\| \tilde{\Lambda}^j (\tilde{\Lambda}^j)^{-1} \right\|_F = \sqrt{\sum_{i=1}^N \sigma_i^2} \quad (3)$$

Where $\|A\|_F$ and σ_i are the frobenius norm of matrix A and i th eigenvalue of $\tilde{\Lambda}^j (\tilde{\Lambda}^j)^{-1}$ respectively. In fact (Γ) measures the maximum dispersion of a point, which transformed with operator $\tilde{\Lambda}^j$ and again transformed back to the first space with $(\tilde{\Lambda}^j)^{-1}$ insted of $(\tilde{\Lambda}^j)^{-1}$ (Which measures How much $\tilde{\Lambda}^j$ is a good approximation of $(\tilde{\Lambda}^j)^{-1}$). In other words diagonal $\tilde{\Lambda}_i^j$ leads to $\Gamma = N$, and this means window j would be a stationary window otherwise we have $\Gamma > N$. Another expression is, since we can rewrite this measure as right equality, Γ measures norm of new eigenvalues due to diagonal approximation of $\tilde{\Lambda}^j$ by $\tilde{\Lambda}^j$, more changes in eigenvalues lead to more non-stationarity situation.

Since each time sample belongs to multiple windows, we construct our non-stationarity filter $\tilde{\Gamma}$ as a average of Γ over all windows contain each time sample. Afterward for extracting stationary and non-stationary parts, each EEG channel could be filtered by multiplying it by $\tilde{\Gamma}$.

First row of Figure ?? depicts four sample EEGs of different subjects and classes and associated non-stationary (over time) measures $\tilde{\Gamma}$ are shown in second row of Figure ?. Non-stationary part of EEGs could be extracted by multiplying $\tilde{\Gamma}$ as shown in third row.

As expected, the peaks in the $\tilde{\Gamma}$ curve appear to match some patterns that appear in the EEG signals. The measure of deviation to a stationary behavior on a graph moments in an EEG signal that we expected to be characteristic of a particular activity.

A. Using this method before classification

For the aim of comparision we reproduced results of [57]. So EEGs are preprocessed as described in II and For each EEG trial, we compute non-stationarity measure over time and filter that EEG by multipling it by this measure. We used CSP followed by RBF-SVM as classifier, with cost and kernel parameters selected from $\{0.5, 1, 2, 4, 8\}$ and $\{0.0625, 0.125, .25, 0.5, 1, 2\}$, respectively by nested cross validation. Average error results for ten-fold crossvalidation over different window sizes, and $S = 1$ are shown in Table ?. Because of some unknown parameters our result doesn't match exactly with results of [57] but they are in acceptable range with respect to the variance that reported in [57].

B. Effect of window size

Fig ?? depicts Classification error over different choices of window size. As one can see, minima of plots rely in $W_t \in [30, 35]$ for four subject. It seems that this window size matches to nonstationary interval of EEGs (duration of MI).

VI. CONCLUSION

In this paper we introduced a data-driven method for non-stationarity detection. Since it makes no assumption about the input data so our method could be used for detecting non-stationarity in every multivariate time series. Experimental results on EEG data shows that this method could decrease average error. Also it can be followed by any classification method, we just used simple linear SVM as a classifier. Also This method potentially has ability to use as an online preprocessing method, this could be a goal for future works.

REFERENCES

- [1] C. Guger, W. Harkam, C. Hertnaes, and G. Pfurtscheller, "Prosthetic control by an eeg-based brain-computer interface (bci)," in *Proc. aaate 5th european conference for the advancement of assistive technology*. Citeseer, 1999, pp. 3–6.
- [2] N. Birbaumer, "Breaking the silence: brain-computer interfaces (bci) for communication and motor control," *Psychophysiology*, vol. 43, no. 6, pp. 517–532, 2006.
- [3] R. Leeb, C. Keinrath, D. Friedman, C. Guger, R. Scherer, C. Neuper, M. Garau, A. Antley, A. Steed, M. Slater *et al.*, "Walking by thinking: The brainwaves are crucial, not the muscles!" *Presence: Teleoperators and Virtual Environments*, vol. 15, no. 5, pp. 500–514, 2006.
- [4] B. Blankertz, G. Dornhege, M. Krauledat, M. Schröder, J. Williamson, R. Murray-Smith, and K.-R. Müller, "The berlin brain-computer interface presents the novel mental typewriter hex-o-spell." 2006.
- [5] G. Cisolto, S. Pupolin, S. Silvoni, and F. Piccione, "An application of brain computer interface in chronic stroke to improve arm reaching function exploiting operant learning strategy and brain plasticity," in *e-Health Networking, Applications & Services (Healthcom), 2013 IEEE 15th International Conference on*. IEEE, 2013, pp. 280–282.
- [6] C. Fadzal, W. Mansor, and L. Khuan, "Review of brain computer interface application in diagnosing dyslexia," in *Control and System Graduate Research Colloquium (ICSGRC), 2011 IEEE*. IEEE, 2011, pp. 124–128.
- [7] K. LaFleur, K. Cassady, A. Doud, K. Shades, E. Rogin, and B. He, "Quadcopter control in three-dimensional space using a noninvasive motor imagery-based brain-computer interface," *Journal of neural engineering*, vol. 10, no. 4, p. 046003, 2013.
- [8] A. S. Royer, A. J. Doud, M. L. Rose, and B. He, "Eeg control of a virtual helicopter in 3-dimensional space using intelligent control strategies," *IEEE Transactions on neural systems and rehabilitation engineering*, vol. 18, no. 6, pp. 581–589, 2010.

- [9] A. J. Doud, J. P. Lucas, M. T. Pisansky, and B. He, "Continuous three-dimensional control of a virtual helicopter using a motor imagery based brain-computer interface," *PloS one*, vol. 6, no. 10, p. e26322, 2011.
- [10] G. Dornhege, B. Blankertz, G. Curio, and K.-R. Müller, "Boosting bit rates in noninvasive eeg single-trial classifications by feature combination and multiclass paradigms," *IEEE Transactions on Biomedical Engineering*, vol. 51, no. 6, pp. 993–1002, 2004.
- [11] G. N. Garcia, T. Ebrahimi, and J.-M. Vesin, "Support vector eeg classification in the fourier and time-frequency correlation domains," in *Neural Engineering, 2003. Conference Proceedings. First International IEEE EMBS Conference on*. IEEE, 2003, pp. 591–594.
- [12] G. Garcia, T. Ebrahimi, and J.-M. Vesin, "Classification of eeg signals in the ambiguity domain for brain computer interface applications," in *Digital Signal Processing, 2002. DSP 2002. 2002 14th International Conference on*, vol. 1. IEEE, 2002, pp. 301–305.
- [13] G. G. Molina and T. Ebrahimi, "Time-frequency-space kernel for single eeg-trial classification," in *Signal processing symposium*, 2002.
- [14] G. N. G. Molina, T. Ebrahimi, and J.-M. Vesin, "Joint time-frequency-space classification of eeg in a brain-computer interface application," *EURASIP journal on applied signal processing*, vol. 2003, pp. 713–729, 2003.
- [15] L. Qin and B. He, "A wavelet-based time–frequency analysis approach for classification of motor imagery for brain–computer interface applications," *Journal of neural engineering*, vol. 2, no. 4, p. 65, 2005.
- [16] V. Bostanov, "Bci competition 2003-data sets ib and iib: feature extraction from event-related brain potentials with the continuous wavelet transform and the t-value scalogram," *IEEE Transactions on Biomedical engineering*, vol. 51, no. 6, pp. 1057–1061, 2004.
- [17] H. Nai-Jen and R. Palaniappan, "Classification of mental tasks using fixed and adaptive autoregressive models of eeg signals," in *Engineering in Medicine and Biology Society, 2004. IEMBS'04. 26th Annual International Conference of the IEEE*, vol. 1. IEEE, 2004, pp. 507–510.
- [18] D. P. Burke, S. P. Kelly, P. de Chazal, R. B. Reilly, and C. Finucane, "A parametric feature extraction and classification strategy for brain-computer interfacing," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 13, no. 1, pp. 12–17, 2005.
- [19] C. W. Anderson and J. A. Bratman, "Translating thoughts into actions by finding patterns in brainwaves," in *Proceedings of the fourteenth Yale workshop on adaptive and learning systems*, 2008, pp. 1–6.
- [20] A. Erfanian and A. Erfani, "Ica-based classification scheme for eeg-based brain-computer interface: the role of mental practice and concentration skills," in *Engineering in Medicine and Biology Society, 2004. IEMBS'04. 26th Annual International Conference of the IEEE*, vol. 1. IEEE, 2004, pp. 235–238.
- [21] H. Serby, E. Yom-Tov, and G. F. Inbar, "An improved p300-based brain-computer interface," *IEEE Transactions on neural systems and rehabilitation engineering*, vol. 13, no. 1, pp. 89–98, 2005.
- [22] M. Grosse-Wentrup, K. Gramann, and M. Buss, "Adaptive spatial filters with predefined region of interest for eeg based brain-computer interfaces," in *Advances in neural information processing systems*, 2007, pp. 537–544.
- [23] M. Grosse-Wentrup, C. Liefhold, K. Gramann, and M. Buss, "Beam-forming in noninvasive brain-computer interfaces," *IEEE Transactions on Biomedical Engineering*, vol. 56, no. 4, pp. 1209–1219, 2009.
- [24] M. Kawanabe, W. Samek, K.-R. Müller, and C. Vidaurre, "Robust common spatial filters with a maxmin approach," *Neural Computation*, vol. 26, no. 2, pp. 349–376, 2014.
- [25] M. Kawanabe, C. Vidaurre, S. Scholler, and K.-R. Müller, "Robust common spatial filters with a maxmin approach," in *Engineering in Medicine and Biology Society, 2009. EMBC 2009. Annual International Conference of the IEEE*. IEEE, 2009, pp. 2470–2473.
- [26] M. Kawanabe and C. Vidaurre, "Improving bci performance by modified common spatial patterns with robustly averaged covariance matrices," in *World Congress on Medical Physics and Biomedical Engineering, September 7-12, 2009, Munich, Germany*. Springer, 2009, pp. 279–282.
- [27] M. Arvaneh, C. Guan, K. K. Ang, and H. C. Quek, "Spatially sparsed common spatial pattern to improve bci performance," in *Acoustics, Speech and Signal Processing (ICASSP), 2011 IEEE International Conference on*. IEEE, 2011, pp. 2412–2415.
- [28] M. Arvaneh, C. Guan, K. K. Ang, and C. Quek, "Optimizing the channel selection and classification accuracy in eeg-based bci," *IEEE Transactions on Biomedical Engineering*, vol. 58, no. 6, pp. 1865–1873, 2011.
- [29] F. Lotte and C. Guan, "Regularizing common spatial patterns to improve bci designs: unified theory and new algorithms," *IEEE Transactions on biomedical Engineering*, vol. 58, no. 2, pp. 355–362, 2011.
- [30] W. Samek, C. Vidaurre, K.-R. Müller, and M. Kawanabe, "Stationary common spatial patterns for brain–computer interfacing," *Journal of neural engineering*, vol. 9, no. 2, p. 026013, 2012.
- [31] W. Samek, "On robust spatial filtering of eeg in nonstationary environments," *it-Information Technology*, vol. 58, no. 3, pp. 150–154, 2016.
- [32] W. Samek, S. Nakajima, M. Kawanabe, and K.-R. Müller, "On robust parameter estimation in brain–computer interfacing," *Journal of neural engineering*, vol. 14, no. 6, p. 061001, 2017.
- [33] W. Samek, D. Blythe, K.-R. Müller, and M. Kawanabe, "Robust spatial filtering with beta divergence," in *Advances in Neural Information Processing Systems*, 2013, pp. 1007–1015.
- [34] W. Samek, M. Kawanabe, and K.-R. Müller, "Divergence-based framework for common spatial patterns algorithms," *IEEE Reviews in Biomedical Engineering*, vol. 7, pp. 50–72, 2014.
- [35] S. Brandl, K.-R. Müller, and W. Samek, "Robust common spatial patterns based on bhattacharyya distance and gamma divergence," in *Brain-Computer Interface (BCI), 2015 3rd International Winter Conference on*. IEEE, 2015, pp. 1–4.
- [36] W. Samek and M. Kawanabe, "Robust common spatial patterns by minimum divergence covariance estimator," in *Acoustics, Speech and Signal Processing (ICASSP), 2014 IEEE International Conference on*. IEEE, 2014, pp. 2040–2043.
- [37] W. Samek and K.-R. Müller, "Tackling noise, artifacts and nonstationarity in bci with robust divergences," in *Signal Processing Conference (EUSIPCO), 2015 23rd European*. IEEE, 2015, pp. 2741–2745.
- [38] R. Soleymannpour and M. Arvaneh, "Entropy-based eeg time interval selection for improving motor imagery classification," in *Systems, Man, and Cybernetics (SMC), 2016 IEEE International Conference on*. IEEE, 2016, pp. 004034–004039.
- [39] S. Lemm, B. Blankertz, G. Curio, and K.-R. Müller, "Spatio-spectral filters for improving the classification of single trial eeg," *IEEE transactions on biomedical engineering*, vol. 52, no. 9, pp. 1541–1548, 2005.
- [40] G. Dornhege, B. Blankertz, M. Krauledat, F. Losch, G. Curio, and K.-R. Müller, "Combined optimization of spatial and temporal filters for improving brain-computer interfacing," *IEEE transactions on biomedical engineering*, vol. 53, no. 11, pp. 2274–2281, 2006.
- [41] K. K. Ang, Z. Y. Chin, C. Wang, C. Guan, and H. Zhang, "Filter bank common spatial pattern algorithm on bci competition iv datasets 2a and 2b," *Frontiers in neuroscience*, vol. 6, p. 39, 2012.
- [42] H. Zhang, Z. Y. Chin, K. K. Ang, C. Guan, and C. Wang, "Optimum spatio-spectral filtering network for brain-computer interface," *IEEE Transactions on Neural Networks*, vol. 22, no. 1, pp. 52–63, 2011.
- [43] A. S. Aghaei, M. S. Mahanta, and K. N. Plataniotis, "Separable common spatio-spectral patterns for motor imagery bci systems," *IEEE Transactions on Biomedical Engineering*, vol. 63, no. 1, pp. 15–29, 2016.
- [44] L. Goldsberry, W. Huang, N. F. Wymbs, S. T. Grafton, D. S. Bassett, and A. Ribeiro, "Brain signal analytics from graph signal processing perspective," in *Acoustics, Speech and Signal Processing (ICASSP), 2017 IEEE International Conference on*. IEEE, 2017, pp. 851–855.
- [45] W. Huang, L. Goldsberry, N. F. Wymbs, S. T. Grafton, D. S. Bassett, and A. Ribeiro, "Graph frequency analysis of brain signals," *IEEE journal of selected topics in signal processing*, vol. 10, no. 7, pp. 1189–1203, 2016.
- [46] X. Xie, Z. L. Yu, H. Lu, Z. Gu, and Y. Li, "Motor imagery classification based on bilinear sub-manifold learning of symmetric positive-definite matrices," *IEEE Transactions on Neural Systems and Rehabilitation Engineering*, vol. 25, no. 6, pp. 504–516, 2017.
- [47] R. Shafipour, S. Segarra, A. G. Marques, and G. Mateos, "Identifying the topology of undirected networks from diffused non-stationary graph signals," *arXiv preprint arXiv:1801.03862*, 2018.
- [48] J. Mei and J. M. Moura, "Signal processing on graphs: Causal modeling of unstructured data," *IEEE Transactions on Signal Processing*, vol. 65, no. 8, pp. 2077–2092, 2017.
- [49] N. Perraudin, A. Loukas, F. Grassi, and P. Vandergheynst, "Towards stationary time-vertex signal processing," in *Acoustics, Speech and Signal Processing (ICASSP), 2017 IEEE International Conference on*. IEEE, 2017, pp. 3914–3918.
- [50] A. Loukas and N. Perraudin, "Stationary time-vertex signal processing," *arXiv preprint arXiv:1611.00255*, 2016.
- [51] F. Grassi, A. Loukas, N. Perraudin, and B. Ricaud, "A time-vertex signal processing framework," *IEEE Transactions on Signal Processing*, 2017.

- [52] G. Mateos, S. Segarra, and A. G. Marques, "Inference of graph topology," *Cooperative and Graph Signal Processing: Principles and Applications* (PM Djuric and C. Richard, eds.), Amsterdam, Netherlands: Elsevier, 2018.
- [53] N. Lu, T. Li, X. Ren, and H. Miao, "A deep learning scheme for motor imagery classification based on restricted boltzmann machines," *IEEE transactions on neural systems and rehabilitation engineering*, vol. 25, no. 6, pp. 566–576, 2017.
- [54] B. Pasdeloup, V. Gripon, G. Mercier, D. Pastor, and M. G. Rabbat, "Characterization and inference of graph diffusion processes from observations of stationary signals," *IEEE Transactions on Signal and Information Processing over Networks*, 2017.
- [55] D.-T. Pham and J.-F. Cardoso, "Blind separation of instantaneous mixtures of nonstationary sources," *IEEE Transactions on signal processing*, vol. 49, no. 9, pp. 1837–1848, 2001.
- [56] M. Moakher, "A differential geometric approach to the geometric mean of symmetric positive-definite matrices," *SIAM Journal on Matrix Analysis and Applications*, vol. 26, no. 3, pp. 735–747, 2005.
- [57] J. Meng, L. Yao, X. Sheng, D. Zhang, and X. Zhu, "Simultaneously optimizing spatial spectral features based on mutual information for eeg classification," *IEEE transactions on biomedical engineering*, vol. 62, no. 1, pp. 227–240, 2015.
- [58] X. Pennec, P. Fillard, and N. Ayache, "A riemannian framework for tensor computing," *International Journal of computer vision*, vol. 66, no. 1, pp. 41–66, 2006.
- [59] J. Angulo, "Structure tensor image filtering using riemannian l^1 and l^∞ center-of-mass," *Image Analysis & Stereology*, vol. 33, no. 2, pp. 95–105, 2014.
- [60] A. Davoudi, S. S. Ghidary, and K. Sadatnejad, "Dimensionality reduction based on distance preservation to local mean for symmetric positive definite matrices and its application in brain–computer interfaces," *Journal of neural engineering*, vol. 14, no. 3, p. 036019, 2017.
- [61] L. M. Machado and F. S. Leite, "Fitting smooth paths on riemannian manifolds," 2004.
- [62] P. C. Petrantonakis and I. Kompatsiaris, "Detection of mental task related activity in nirs-bci systems using dirichlet energy over graphs."

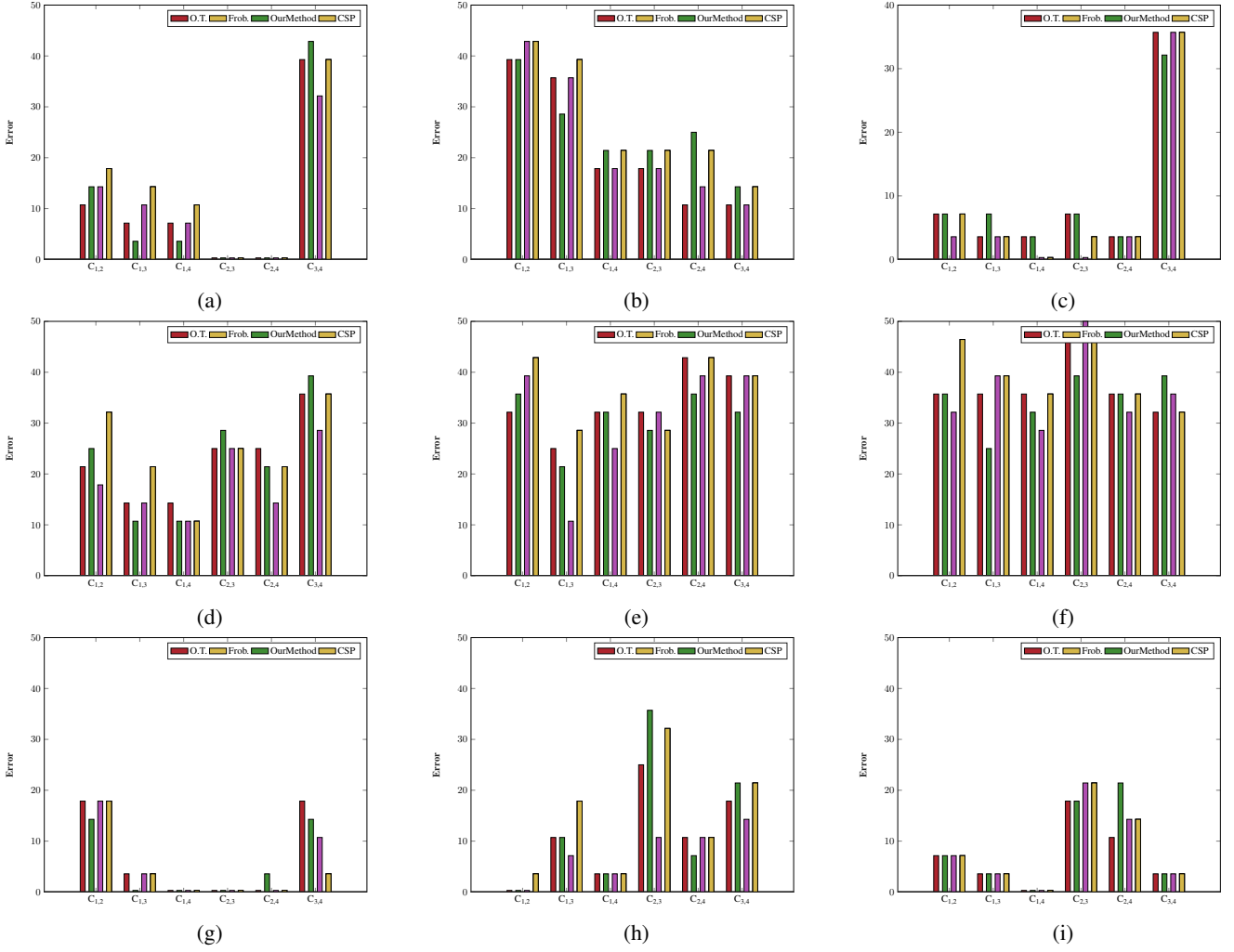


Figure 1: SVM error over different classes for best window size selection (BCI competition IV data set IIa).

	C1_C2		C1_C3		C1_C4		C2_C3		C2_C4		C3_C4		Average			C1_C2	
	CSP	Our	CSP	Our	CSP	Our	CSP	Our	CSP	Our	CSP	Our	CSP	Our		CSP	Our
S01	17.86	14.28	14.28	10.71	10.71	7.14	0	0	0	0	39.28	32.14	13.69	10.71	aa	25	28.57
S02	42.86	42.86	21.43	35.71	21.43	17.86	21.43	17.86	21.43	14.28	14.28	10.71	26.78	23.21	al	0	3.57
S03	7.14	3.57	3.57	3.57	0	0	3.57	0	3.57	3.57	35.7	35.41	8.92	7.73	av	25	25
S04	32.14	17.86	21.43	14.28	10.71	10.71	25	25	21.43	14.28	35.71	28.57	24.4	18.45	aw	0	0
S05	42.86	39.28	28.57	10.71	35.71	25	28.57	32.14	42.86	39.28	39.28	39.28	36.31	30.95	ay	10.71	3.57
S06	46.43	32.14	39.28	39.28	35.71	28.57	46.43	50	35.71	32.14	32.14	35.71	39.28	36.31	Average	12.14	12.14
S07	17.86	17.86	3.57	3.57	0	0	0	0	0	0	3.57	10.71	4.17	5.36			
S08	3.57	0	17.86	7.14	3.57	3.57	32.14	10.71	10.71	10.71	21.43	14.28	14.88	7.74			
S09	7.14	7.14	3.57	3.57	0	0	21.43	21.43	14.28	3.57	3.57	3.57	8.33	8.33			
Average	24.20	19.44	19.04	14.28	13.09	10.31	19.84	17.46	16.66	14.28	25	23.41					

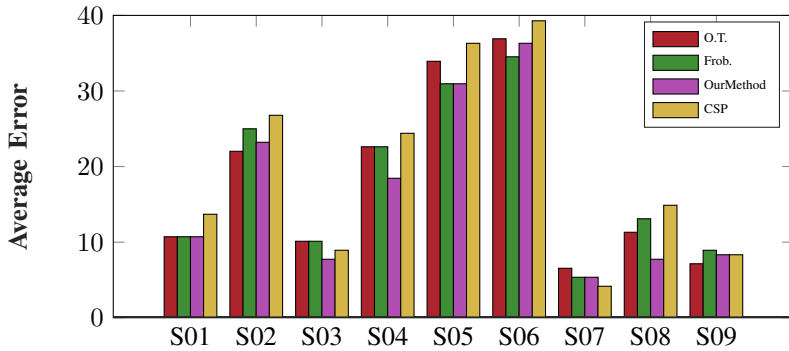


Figure 2: Average SVM error overall classes for each subject: (BCI competition IV data set IIa).

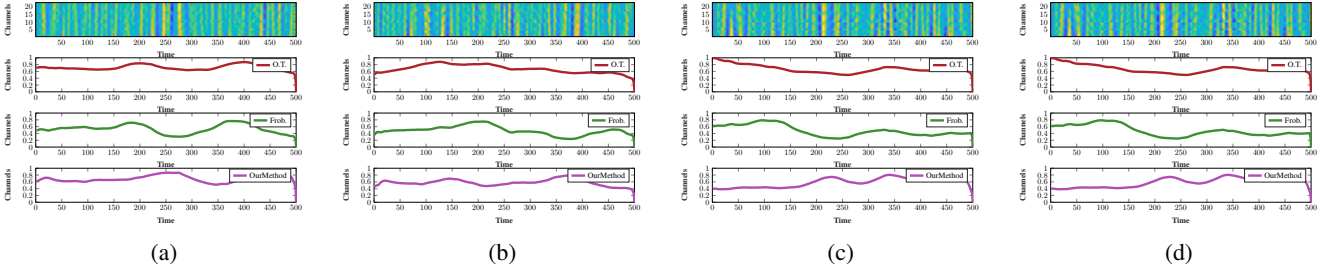


Figure 3: Different Non-stationary measures for random selected trials (50,114,150,200) from subject A01 (BCI competition IV data set IIa)